Fiscal Year:	FY 2015	**Task Last Updated:**	FY 10/30/2014
PI Name:	Olson, Sandra Ph.D.		
Project Title:	Correlation of Normal Gravity Aerospace Materials Flammability Data with Flammability in Microgravity Environments		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Combustion science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Organization Name:	NASA Glenn Research Center		
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City:	Cleveland	**State:**	OH
Zip Code:	44135	**Congressional District:**	9
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2009 Combustion Science NNH09ZTT001N
Start Date:	01/01/2010	**End Date:**	12/31/2019
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	1
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:	Urban, David	**Contact Phone:**	216-433-2835
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Flight Program:			
Flight Assignment:	NOTE: Extended to 12/31/2019, per PI (Ed., 1/26/2015) NOTE: Extended to 12/31/2014, per PI (Ed., 10/17/13) NOTE: Extended to 12/31/2013, per PI (Ed., 10/10/2012)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Hirsch, David (NTEC, Jacobs Engineering)		
Grant/Contract No.:	Internal Project		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NASA STD 6001 Test 1 is the major method used to evaluate flammability of materials intended for use in habitable environments of U.S. spacecraft. The method is an upward flame propagation test initiated in a stagnant environment and using a well-defined igniter flame at the bottom of a vertically mounted sample. A material passes this test if the vertical burn length is less than 15.2 cm (6 inches) and there is no evidence of transfer of burning debris. The test is conducted in the most severe flaming combustion environment (oxygen concentration, pressure) expected in the spacecraft, currently 30% oxygen at 10.2 psia. Recent research has shown that current normal gravity materials flammability tests do not correlate with flammability in ventilated, micro-, or reduced-gravity conditions. The materials selection for spacecraft is based on the assumption of commonality between ground test flammability results and spacecraft environments, which does not appear to be valid. To better understand the actual normal gravity flammability limits, a modified Test 1 protocol has been proposed that, as an alternative to qualifying materials as pass/fail in the worst-expected environments, measures the actual upward flammability limit (akin to the Limiting Oxygen Index (LOI)) for the material so that a more accurate assessment of the margin of safety of the material in the real use environment can be made. The method allows the option of selecting better or best materials based on the margin of safety, as opposed to what would be considered just "passing" from a flammability point of view. A research program is underway to correlate normal gravity flammability test data with data under ventilated reduced gravity or microgravity conditions. The program will determine materials suitability for microgravity and reduced gravity testing by evaluating oxygen concentration thresholds under various total pressures in normal gravity tests and in spacecraft-specific environments (ventilated microgravity or reduced partial gravity). The normal gravity data will be correlated with flammability data in microgravity environments and the safety factors involved when selecting spacecraft materials based on normal gravity testing will be evaluated. Rigorous correlations between ground test flammability data and data in ventilated spacecraft environments would allow selection of materials with increased fire resistance, and thus decrease the fire risk in space systems. The operational flexibility will also increase since the safety factors will be assessable.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The identification of the fundamental limit in microgravity and its comparison with the normal gravity flammability limit will help us understand flammability in different configurations on Earth. It will improve the safety of astronauts in spaceflight so they can return safely to Earth.
Task Progress:	This year the BASS-II precursor experiments were conducted aboard ISS using the Microgravity Science Glovebox and the BASS hardware. For this project (SMµRF), acrylic cylinders in 3 diameters were tested in a concurrent (stagnation) flow to determine flammability limits as a function of flow and ambient oxygen level. BASS-II Ops were completed from February to early August, 2014. Science data analysis is underway. A total of 24 SMµRF tests were conducted during BASS-II. Some samples were burned more than once. There are some samples that could be burned further, should future operational time become available. The Principal Investigator is busy analyzing the results of BASS-II, in preparation for a Requirements Definition Review in Dec. 2014. For the first time on ISS, BASS-II utilized MSG working volume vitiation with GN2. We developed a perfectly stirred reactor model to determine the GN2 flow time and flow rate to obtain the desired reduced oxygen concentration in the working volume for the tests. This worked out extremely well for operations, and added a new vital variable, ambient oxygen level, to our test matrices. We obtained blowoff and quenching extinction limits to define the flammability boundary for PMMA as a function of flow and oxygen concentration. The low flow boundary is a surprisingly steep threshold at very low flows, whereas blowoff is a more gradual slope. The 1g blowoff for the ¼” rod is at 16% oxygen, whereas we obtained steady spreading at 13.6% oxygen in BASS-II. Thus the material has a negative Oxygen Margin of Safety of 2.4% O2, and the normal gravity flammability limits should be de-rated by at least that much. The minimum in the flammability boundary is at very low flows, on the order of diffusive flows. This minimum occurs in what is termed here the ‘Goldilocks’ zone of flow, where the oxygen flow is not too little to suffocate the flame, provides just enough heat release to sustain the flame but the flow is not so strong that it sweeps the fuel out of the hot flame zone before the reaction of Equation 1 can occur. As ambient oxygen is reduced, the flame temperatures are reduced, which slows down the reaction time. Thus the flow needed to blow off the flame becomes slower as oxygen decreases and the flame temperature is reduced. Each test provides a wealth of information in addition to the extinction limits. This data includes flame size, standoff distance, spread rate, bubble layer thickness, tip regression shape, pre-extinction flame oscillations, and vapor jetting frequency and sample interactions. This data is still being analyzed. Ground-based drop tower work and lab experiments have been conducted to develop an illumination technique to image the bubble layer in depth while still imaging the flame shape. The Science Requirements Document for the flight experiment is being updated to reflect the knowledge gained from the BASS-II tests and the ground-based testing. A Requirements Definition Review is scheduled for December, 2014.
Bibliography Type:	Description: (Last Updated: 01/12/2026)
Abstracts for Journals and Proceedings	Olson SL, Ferkul PV, Bhattacharjee S, Miller FJ, Fernandez-Pello AC, T'ien JS. "Overview of Burning and Suppression of Solids – II Fire Safety Investigation for the Microgravity Science Glovebox." Presented at International Workshop for ISS Material Flammability Standard Project, JAXA Tsukuba Space Center, Tsukuba, Japan, January 23-24, 2014. International Workshop for ISS Material Flammability Standard Project, JAXA Tsukuba Space Center, Tsukuba, Japan, January 23-24, 2014. , Jan-2014

Abstracts for Journals and Proceedings	Olson SL. "Development of an ISS Experiment to Study Spacecraft Materials Flammability Evaluation Test Methods." Presented at the 30th Annual Meeting of the American Society for Gravitational and Space Research, Pasadena, CA, October 22-26, 2014. 30th Annual Meeting of the American Society for Gravitational and Space Research, Pasadena, CA, October 22-26, 2014. , Oct-2014
Papers from Meeting Proceedings	Olson SL, Beeson H, Fernandez-Pello AC. "Applying Flammability Limit Probabilities and the Normoxic Upward Limiting Pressure Concept to NASA STD-6001 Test 1." 44th International Conference on Environmental Systems, Tucson, Arizona, July 13-17, 2014. 44th International Conference on Environmental Systems, Tucson, Arizona, July 13-17, 2014. Paper number ICES-2014-179. http://hdl.handle.net/2346/59738 ; accessed 11/4/14. , Jul-2014